



Title: AI, Surveillance, and the Fight for Freedom in Authoritarian Regimes

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Abstract

This thesis investigates the role of AI in authoritarian repression with a case study of the Islamic Republic of Iran. While global discussions regarding AI have mainly focused on its impact on democratic societies, this study refocuses the lens, examining how autocratic governments employ AI technologies for surveillance, chill, and repress dissent. Critical analysis of the politics and practices of governments, companies, researchers, and AI development in the industrialized North and the Global South (including the politics and practices of oppression, resistance, and liberation in the latter) Inspired through frameworks well established in the Frankfurt School, the research make use of thematic analysis to conduce the collected qualitative data from in-depth interviews with AI experts and Iranian activists. The findings illustrate a complex digital repression system where AI, for instance, facial recognition, predictive policing, and algorithmic content moderation, is strategically used to consolidate authoritarian rule. They described surveillance in both online and offline spaces, algorithmic censorship on social media, and laws that enshrine repression under the pretext of protecting users. Yet under such conditions of inequality and unfreedom, the study notes various forms of resistance from below, from encrypted communications and digital literacy training to adversarial techniques for AI surveillance. The thesis maintains that AI, in authoritarian spaces such as Iran, is not a neutral tool but is an infrastructural technology laden with politics that augments power imbalances. By focusing on the lived experiences of affected communities, this work adds to the growing discussion surrounding digital authoritarianism by providing empirical insight followed by normative reflection for policymaking and research on AI governance, human rights, peace, and development.

Keywords: Artificial Intelligence (AI), Authoritarianism, Digital Repression, Surveillance, Freedom of Speech, Activism, Human Rights



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List of Abbreviations

Abbreviation	Full Term
AI	Artificial Intelligence
STS	Science and Technology Studies
CT or CT&S	Critical Theory (of Technology and Science)
HCAI	Human-Centered Explainable Artificial Intelligence
IRGC	Islamic Revolutionary Guard Corps
VPN	Virtual Private Network
NLP	Natural Language Processing
OECD	Organisation for Economic Co-operation and Development
IEEE	Institute of Electrical and Electronics Engineers
UPCOS	User Protection and Core Online Services (Bill)
UN	United Nations
EU	European Union



FLI	Future of Life Institute
VoA	Voice of America



1. Introduction

The rapidly multiplying uses of artificial intelligence (AI) are remaking not just the economy, but on a more sinister note, the tools of contemporary authoritarian rule (Feldstein, 2021). According to the European Parliamentary Technology Assessment Network (2024) although academics pay a lot of attention to how AI threatens liberal democratic institutions, undermining transparency, trust, and freedom of expression, they have not paid nearly enough attention to how, for non-democratic regimes, AI can be abused to harden authoritarianism and quash civic resistance. This thesis explores how authoritarian governments, with a specific interest in the Islamic Republic of Iran, use AI technologies to spy on activists. Despite an increasing international focus on the moral implications of AI, a gap remains in research that addresses the ways that these technologies are weaponized against political dissent in authoritarian environments. The Iranian state's reliance on facial recognition, online content moderation, and surveillance apparatus has increased in recent years, particularly in reaction to movements like "Woman, Life, Freedom." This raises important questions about the transforming ties between the development of technology and repressive control.

The research uses Iran as a case study to explore architectural elements that allow the use of AI "technologies" to suppress voices. It also aims to engage in more general debates around the theme of digital authoritarianism, offering analyses that could help shape policy responses and civil society strategies to ensure that democratic values are upheld in the digital sphere.

Finally, my study contributes to peace and development studies by linking international debates about AI governance to local struggles for democratic voice. This study not only fills a critical empirical and theoretical gap but also challenges the dominant narrative of AI as a neutral or predetermined force into a contested and politically charged tool, especially where repression and invention coincide.



1.1 Research Context

The use of AI by authoritarian states has significantly changed the dynamics of digital repression during the last decade (Ünver, 2024). From facial recognition and predictive policing to algorithmic censorship and surveillance based on big data, they are no longer neutral, they are integrated into the infrastructures of repressive states. Although attention has mostly been drawn to AI's disruption of democratic systems in the Global North, much less has been paid to the systematic deployment of AI by non-democratic regimes to crack down on civic activists and dissent.

This study aims to fill this gap through a study of the Islamic Republic of Iran, where AI is deployed as part of a comprehensive system of political control. One of the most vivid examples is the regime's use of facial recognition technologies to enforce mandatory hijab regulations surveillance cameras with AI recognition tools have been placed in crowded places and in public transport to enforce women's "proper" wearing of hijab and culprits can now face fine, including confiscation of car or even being arrested (Amnesty International, 2024). The case shows not just how AI is being used to police dissent, but also to compel ideological orthodoxy.

This research diverges from comparative liberal-democratic research on AI impact, such as conversations around trust erosion or silencing of speech, to examine how the authoritarian regime is using AI as a weapon of moral and ideological control. This thesis employs and further develops the new interdisciplinary field of critical data studies to frame digital infrastructures as politically charged systems that have the potential to sustain contemporary modes of oppression. Drawing on expert interviews and Iranian activist testimonies, as well as NGO reporting and legal analysis, the research offers a detailed account of how AI-facilitated repression plays out in actual authoritarian contexts.



1.1.1 Research Questions:

- How does an authoritarian regime in Iran deploy artificial intelligence to suppress civil activism?
- How does AI impact the behavior, communication, and security of political and civil activists in Iran?

1.2 Justification of the study

First, unlike most studies that analysed authoritarian digital repression from afar, and I will mention in the literature review, this research presents a story from within Iran's digital authoritarianism. In addition to my first-hand experiences, as an Iranian activist, journalist, and former political prisoner provide access to a depth of interpretation that few academic studies can achieve. Most of the literature in this area relies on public data and legal analyses. On the other hand, this research adds original empirical data based on interviews with actors, activists, and other AI researchers that further illuminate this process's human and human-related aspects. This research uses critical theory to expose the power asymmetries encoded in AI infrastructures and to counter technological solutionist narratives. Also, as AI technologies become more of a tool for control than neutrality, especially in areas where accountability is not working correctly, peace and development studies need to pay more attention to the digital dimension of repression. This study attempts to contribute to this subfield by linking political theory, ethics, and human security.

While AI can support democratic participation in some countries where it has already been implemented, in others, especially non-democratic or hybrid states, it raises urgent questions of repression, civil liberties, and civic agency (Bukhari & Anwar, 2025). Focusing on Iranian activists, this study examines how AI reinforces authoritarian power but does not necessarily empower citizens through structural and dynamic mechanisms. Furthermore, it is at the



intersection of digital governance, democracy studies, and peace and development. By examining how AI can be used against activists, this dissertation contributes to a broader understanding of technological systems that enable peace and civic inclusion or exacerbate tensions and deprivation.

1.3 Definitions: AI, freedom of speech, authoritarian government

First, we will specify the definitions we intend to use here to ensure we are on the same page and have a common understanding of the words used in this thesis.

1.3.1 AI

Artificial Intelligence (AI) is a simulation of human intelligence that operates by machines, typically computer systems, such as learning, reasoning, problem-solving, perception, and language comprehension (Russell & Norvig, 2021). AI is "the study of agents located in an environment that receives precepts from the environment and performs actions" (Russell & Norvig, 2021, p. 1). These agents are typically developed to resemble cognitive processes like learning and adaptation (Nilsson, 2009), and they function based on a mix of algorithms, data, and computational horsepower (Jordan & Mitchell, 2015).

1.3.2 Freedom of Speech

Freedom of Speech is the right of people to express their opinions and ideas without fear of punishment or government interference, so long as those words do not provoke violence or hate, drama for the sake of drama, or perpetuate racism, sexism, and discrimination (UN General Assembly, 1948, Article 19). It underlies the development of democratic politics and democratic societies more generally by providing for freedom of expression and association, open communication and exchange of ideas, and the holding of power to account (Barendt, 2005; Mill, 1859/2003). This right shall include freedom to hold opinions and to receive and impart information and



ideas through any media regardless of frontiers (UN General Assembly, 1948, Article 19).

Freedom of speech is not a blank check; restrictions can be made to protect national security, safeguard public order, or protect the rights of others (Penn Today, 2012). However, in liberal democratic theory, it is regarded as a core component of personal autonomy, the pursuit of truth, and the proper operation of democracy (Dworkin, 1996; Waldron, 2012).

find

Freedom of expression index, 1789 to 2024

Based on the expert estimates and index by V-Dem. It captures the extent to which people can voice their views and the media can present different political perspectives. It ranges from 0 to 1 (most free).



Source: ourworldindata.org

According to the Free of expression report 2024 Iran is consistently rated one of the worst countries in the world for press freedom. Reporters are harassed, arrested, and jailed for covering sensitive topics. Internet freedom is also restricted as the government objects by blocking websites and some social media. During the unrest, the authorities have slowed internet speeds or cut off access to impede protesters' communication (Penn Today, 2012).

1.3.3 Authoritarian Government

A dictatorial regime is a political system having relatively only a small degree of political pluralism, without meaningful or competitive democratic



competition, in which power is concentrated in the leader or leadership not constitutionally responsible to the body of the public (Linz, 2000; Levitsky & Way, 2010). Authoritarian rules are the opposite of democratic rules; rather than promote participation, they usually repress political opponents, reduce civil liberties (including freedom of speech, freedom of assembly, and free media), and use force and surveillance to control their population (Linz, 2000; Levitsky & Way, 2010).

According to Levitsky, S., & Way, L. A. (2010), in a normal Authoritarian Government, power is concentrated in a single ruler with a handful of elite friends and with few institutions to check and balance the ruler and minority. These regimes frequently evade, then seek to corrupt, the rule of law, undercutting the independence of the judiciary and the separation of powers. The judiciary is often used not for justice but to give legitimacy to the actions of the regime and to represent the opposition. If elections occur, they are usually closely supervised or orchestrated.

In authoritarian regimes, the opposition is traditionally crushed through censorship, unlawful detention, violence, and coercion. Civil society and independent media are heavily controlled, if not entirely suppressed, and little space is left for public oversight and free speech (Schenkkan & Linzer, 2021). The regime regularly resorts to repressive state tools, the military, the police, and intelligence agencies to maintain a grip on power and to punish challenges. These features enable the ruler to keep a grip on power and restrict the opportunity for civil participation and democratic rule (Schenkkan & Linzer, 2021).

Iran is usually portrayed as an authoritarian, but not totalitarian state based on its contemporary history. Although its findings share some similarities with those of other studies of authoritarianism (e.g., centralization of power, repression of opposition, and control of information), they do not exactly conform to the models of totalitarianism formulated by scholars such as Juan José Linz (Linz, 2000).



Linz (2000) describes totalitarian regimes as those with a government that intends to control all aspects of public and private life, including the public ideology, a mass party led by a supremely dictatorial leader, a monopoly of means of communication and terror, and a monopolistic control over the anti-democratic economy. By contrast, in Iran's authoritarian polity, there is room for minimal pluralism and civil society activity, even if it is restricted. For example, Iran does have periodic, approximating to regular, elections, and it does happen to showcase some semblance of a multiplicity of state-based political factions, but the Guardian Council pre-screens the candidates and will disqualify (exclude from political contention) those found not to be appropriately "suited" to the task of genuine political representation/competition (Human Rights Watch, 2005). Also, the philosophy behind the Iranian regime, this specific interpretation of a religion, Shias Islam, informs their laws and the way they govern. However, it is not an ideology not in the kind of universal terms we would understand as Marxism-Leninism in the Soviet Union or Maoism in China.

2. Literature review

The intersection of artificial intelligence (AI), free speech, and authoritarian governance has become a critical area of study, particularly as AI technologies increasingly shape civic space and modes of political repression. Scholarship highlights that while AI can empower activism and democratize information, it facilitates unprecedented surveillance and control when deployed by authoritarian states.

2.1 AI, Free Speech, and Authoritarian Repression

Authoritarian regims have a history of hijacking digital media for control purposes (Burgers & Robinson, 2016), fueling concepts like “networked authoritarianism” (MacKinnon, 2011), which refute the neutrality of technology (Samarakoon et al., 2017). Groundbreaking research has shown that the architecture of digital technologies are not necessarily conducive for



democratization, but infiltrated by authoritarian structures of control and surveillance (Deibert, 2013; Mozorov, 2011).

The surveillance agency is at the heart of authoritarianism; a tool for collecting intelligence on threats, quelling protests, and creating consent (Schedler, 2013; Richards, 2013). This ‘chilling effect’ will be sustained by informatics of the kind that models reality and in the process solidifies regime resilience through fear and control. Yet recent scholarship advances on this approach by showing that digital repression spans governance, political science, and international relations (Feldstein, 2021; Guriev & Treisman, 2019; Keremoğlu & Weidmann, 2020).

There is an increasing interest in AI in such activities.” Cevallos (2025) outlines how tools such as predictive policing and facial recognition help authoritarian regimes to stifle activism more effectively. Akbari (2022) demonstrates how ‘smart city’ infrastructures naturalise surveillance as modernisation, incorporating control into everyday life. These technologies are being used with little ethical scrutiny, particularly in jurisdictions lacking transparency standards (Elkin-Koren & Perel, 2023; Ridley, 2025).

In totalitarian locations like Iran, AIs go beyond just civilian surveillance in ideologies too – like gendered control through hijab recognition systems (Amnesty International, 2024). These instruments do not just discipline behavior, but “institutionalize theocratic decrees through public” channels, which makes the Iranian model different from that of secular tyrannies. However, in many cases this localized application is overlooked in most studies, which tend to adopt more abstract or liberal democratic perspectives (Duberry, 2022; Filgueiras, 2022).

The literature also describes tactics for resistance to these technologies, including explainability activism and civic audits (Krafft et al., 2021). But those are predicated on a legal system and civil society capacity that simply doesn’t exist in closed regimes like Iran. There, opacity is not a bug, but a feature, intentionally designed to prevent accountability.



Analogue cases (e.g., China’s Great Firewall, or Russia’s project for a sovereign Internet) demonstrate how state-driven digital governance models emerge in response to perceived threat (MacKinnon, 2008; Soldatov & Borogan, 2015). These models are highly reflective of Iran’s own efforts at internet infrastructure control and mobilization repression, such as its reaction to the 2009 Green Movement and the “Woman, Life, Freedom” protests (Golkar, 2011; Akbari & Gabdulhakov, 2019).

The notion of socio-technical imaginaries (Jasanoff & Kim, 2015) might be helpful when understanding the ways in which authoritarian regimes give voice to their utopia of order and efficiency through the use of AI surveillance. In Iran, this vision manifests in a blending of security, ideology, and morality—a theocratic approach to digital authoritarianism.

This study furthers this literature by examining how the hybrid regime in Iran localizes AI in religious and political repression—going beyond generic AI ethics talk to spotlight the material implications for activists. It also speaks to an increasingly mature but as yet underdeveloped literature on authoritarian AI that provides an empirically based, theoretically informed account of one of the most closed digital states.

2.2 Islamic regime in Iran as a totalitarian government

Iran is shaped by and mediated through a complex nexus of social, cultural, political, and technical factors, most notably within the context of Islamic regulatory frameworks.

Building on this context, In Iran, take-up of the internet has been one of the most rapid in the MENA region—only surpassed by Israel. In 2015, the number of Iranians who had online access was approximately 47 million, some 57.2% of its 80 million citizens.

However, the growing use of the internet has been accompanied by a notable increase in users attempting to bypass state-imposed restrictions. According to Iranian government data from 2015, more than 23.5 million young people



in the country relied on virtual private networks (VPNs) or proxy servers to evade online censorship.

This widespread circumvention highlights a critical tension between digital adoption and state control. Moreover, it is widely assumed that social media significantly facilitated post-election protests in 2009. In this sense, the Persian Summer, as we term it, can be understood as a precursor to the Arab Spring, which began in late 2010 (Akbari and Gabdulhakov, 2019).

Such events underscore the regime's anxieties about digital mobilization. According to the latest Freedom on the Net report for 2023, Iran, alongside Myanmar and China, has the worst situation in terms of users' free access to the Internet among 70 countries. This index includes components such as violations of user rights, barriers to access, and restrictions on content. Iran has been ranked as the worst country in terms of respecting citizens' rights and fourth in terms of Internet content restriction (censorship) among the countries surveyed in the sub-indices of this survey.

This deeply restricted environment reflects broader patterns of governance in Iran. Since the early 2000s, the widespread adoption of the internet in Iran has enabled the public to circumvent state censorship, prompting concerns among the ruling authorities who perceived it as a threat to their control (Rahimi, 2003). In reaction, the government has implemented a range of censorship strategies aimed at asserting control over both digital and physical spaces (Kawerau et al., 2022).

As Polyakova and Meserole (2019) define digital authoritarianism as 'the use of digital information technology by authoritarian regimes to monitor, suppress, and manipulate domestic and foreign populations. More specifically, digital authoritarianism restricts the flow of information and attempts to disconnect online activities from offline developments.

The 2009 Green Movement marked a pivotal moment in highlighting the intersection of digital activism and physical dissent against the regime. Iranian protesters effectively utilized social media platforms like Facebook



and Twitter to coordinate street demonstrations in response to allegations of electoral fraud (Golkar, 2011). Despite increased restrictions, access to social media has offered new opportunities for voices of dissent to be amplified (Khiabany and Zayani 2019).

2.3 Comparative Evaluation of the Literature

In parallel with Iran's digital trajectory, a treasure trove of literature on AI and authoritarianism elucidates how states co-opt digital technologies to surveil, shape, and stifle civic agency. Nevertheless, a cross-sectional analysis of theory and model contributions hints at strengths and weaknesses in the existing literature. This sub-section critically reviews the literature about AI and then locates this thesis within the literature before explaining a new knowledge contribution, placing AI's use in the Islamic Republic of Iran in context.

Foundational studies laid the groundwork for this analysis. The conceptualizations of networked authoritarianism and state capture of digital infrastructures were instrumentalized in many early investigations (MacKinnon 2011; Deibert 2013). These works opened the door for more recent investigations of AI-facilitated repression.

In this regard, researchers such as Akbari (2022) and Vanolo (2014) focus on urban infrastructures as a key mechanism for embedding surveillance into the everyday, illustrating how “smart cities” work to naturalize authoritarian control. On a different note, Duberry (2022) and Ridley (2025) emphasize algorithmic opacity and its impact on trust and public contestation, exposing the epistemic risks of information and discourse management, which underpins the utilization of black-boxed AI systems.

While these contributions are valuable, they tend to treat authoritarianism as a general or abstract category. They tend to underplay how particular regimes defined by ideologies, systems of governance, and cultural habits take up AI in heavily localized manners. For example, Duberry and Ridley concentrate on democratic or semi-democratic societies and do not explain how



authoritarian states without any mechanisms of accountability utilize AI in the most coercive ways.

This is where this thesis intervenes. It fills that gap through an analysis of Iran's hybrid theocratic-authoritarian regime in which AI is a tool not only for surveillance but also for religious enforcement (e.g., hijab recognition systems), gendered control, and transnational repression. Furthermore, a substantial proportion of the literature focuses on legal/ethical/technical responses to AI repression, such as HCXAI, civic audits, and challenge ability (Krafft et al., 2021; Elkin-Koren & Perel, 2023).

However, such tactics assume a civic space and rule of law that do not exist in authoritarian regimes such as Iran. Consequently, our research challenges these assumptions, demonstrating that, in closed regimes, AI's opacity is not a bug to be fixed but a feature carefully engineered to sidestep accountability.

3. Theoretical Framework

This study employs Critical Theory to examine how AI technology is a tool for political control in authoritarian regimes and how activists respond to or resist it. These theories provide both a normative and an analytical tool for critically examining the power structures that shape digital infrastructures.

This dual framework links macro-structural critique with grounded social practices to build a comprehensive understanding of systemic control and the possibility of resistance.

3.1 Critical theory

Critical theory emerged in the 1920s and 1930s through the work of Western European Marxist thinkers associated with the Frankfurt School. It is grounded in six central concepts: text, meaning, culture, polysemy, hegemony, and ideology. More broadly, critical theory encompasses any approach that aims to critique and transform society in the pursuit of emancipation and empowerment, making the analysis of power a core



element of its framework (Corradetti, 2018). Among its key figures, Herbert Marcuse played a significant role by connecting the Frankfurt School's critique of rationality to a concrete examination of how technology shapes life in advanced industrial societies (Feenberg, 1995). This linkage becomes especially relevant in contemporary digital societies, where AI systems increasingly shape human behaviour and state power.

The Frankfurt School members interpreted and expanded Marxist society into the cultural realm. This concept was then extended to the context of mass communication. One does not have to look very hard to see how radio, film studios, newspapers, magazine publishers, and eventually television businesses have conformed to the capitalist mode of production (Tyson, 2023).

Critical Theory, as it developed in the Frankfurt School, is best understood less as a substantive body of thought and more as a set of methodological aims that distinguish it from many other kinds of social and political theorizing (in both philosophy and the social sciences): It seeks to be (1) self-reflexive, tracing its genealogy from specific social and historical moments, (2) interdisciplinary, drawing on and integrating philosophical analysis with conventions of social theory and empirical social research, (3) materialist, rooting its critiques in social reality and (4) emancipatory, orienting itself to the goal of social criticism. These commitments firmly place the Frankfurt School within the Marxist tradition and the latter's project of supplanting the division between theory and practice in a manner that is critically self-reflective and does not simply passively subsume one into the other (Bohman, Flynn & Celikates, 2021). This theoretical foundation directly informs the methodological approach of this study, which integrates critical inquiry with qualitative fieldwork.

Critical theorists argue that ideologies are neither neutral nor harmless; rather, they serve to legitimize and reinforce existing power hierarchies by shaping how individuals perceive and interpret reality. By critically examining ideology, Critical Theory reveals the underlying mechanisms—



ranging from cultural norms to institutional practices—that perpetuate social inequality (Zuboff, 2019).

This study employs critical theory because it examines how AI tools, including facial recognition, predictive policing, and online moderation, are utilized as instruments of state power in Iran and how these systems perpetuate or reinforce pre-existing inequalities and political oppositions. Ethical frameworks, such as those offered by HCXAI, also tend to ignore the political nature of embedding technology in authoritarian environments.

With origins in sociology and literary criticism, the Frankfurt School and, more broadly, critical theories aim to shed light on the conditions that enslave people and seek “to liberate human beings from the circumstances that enslave them” (Rasmussen, 2001). These theories emerged about social movements, including LGBTIQ+ minorities. In both the broad and narrow senses, critical theory provides the descriptive and normative foundations for social inquiry that aim to reduce domination and enhance freedom in all its dimensions (Bohman, 2021). This emancipatory mission resonates with the goals of activist resistance to AI surveillance, especially in politically repressive environments.

Feenberg believes that the critical theory of technology offers a platform for reconciling many conflicting strands of reflection on technology. We can only make sense of the world through a crucial and empirically oriented approach. The first generation of Critical Theorists called for just such a synthesis of theoretical and empirical approaches (Feenberg, 2005). This study reflects that call by combining theoretical insights with empirical data from interviews with activists and experts.

Feenberg (2005) argues that the die-hard core of the critical theory of technology is not technology or progress as such but rather the spectrum of technological options and trajectories from which any society must make fit choices (e.g. the choice of ink and paper as opposed to computer, over clay...). He argues that modern technology, like medieval cathedrals or the



Great Wall of China, is not politically neutral but instead embodies the values of industrial society and, in particular, of dominant elites who make use of the authority conferred by technical expertise. Against this, Feenberg argues for a cultural criticism that contests values built into technology systems. This kind of criticism allows us to imagine an industrial society of a different kind based on other principles. Whilst this vision is appealing, he claims, it calls for a kind of critical rationality that can think about the larger social and cultural context of technology.

By examining the specific values embedded in Iranian AI systems, this study reveals how technical artifacts reinforce hegemonic cultural and political norms. Rather than celebrating the increasingly rapid development of technology, the Frankfurt School and later critical theorists began with the counterintuitive idea that the human potential for self-determination and rational organization appears to diminish rather than increase as scientific-technological progress advances. The paradox that critical theorists diagnose originates in our dependence on reason and rationality. Their ideas also speak directly towards present-day discussions of AI and ethics, and especially towards contemporary anxieties about technology's potential for domination and control, foreboding currents in criticism of the outcomes wrought by Big Data and AI (Zuboff, 2019).

Critical theory calls into question the neutrality of technology and proposes that technological rationality has become a new form of political rationality. The interests and values of the ruling classes are inscribed in built-up technological systems. This dominant form of technical rationality is not simply an expression of ideology or a natural technical necessity but has its actuality in the intersection of ideology and technics. At the moment, both exist primarily to control human beings and resources in accordance with elites. Feenberg (2005) terms this the working of “technical codes”—the (frequently unnoticed) rules and systems by which technologies internalize and articulate hegemonic values to enable the everyday exercise of power and social control. This insight is essential to understanding how Iran's AI



systems operationalize control through seemingly technical but deeply political design choices.

3.2 why Critical theory

In this research, we examine how authoritarian states specifically, Iran use AI to suppress dissent. This requires a qualitative framework that goes beyond the simple display of technology, and a critical approach allows for this. The prior section established the foundations of critical theory; here, its specific relevance to the research problem is outlined.

There are five main reasons why critical theory is the best theoretical framework for this study. First, it focuses on power and dominance, which makes it very useful to analyze how authoritarian regimes like Iran embed control in AI infrastructures such as facial recognition and predictive policing. Second, it has both an emancipatory and an activist orientation, which is consistent with the idea of this thesis, not simply to identify repression but to find points of resistance and liberation. Third, critical theory views technology as an ideological rather than a neutral concept, allowing for examining how AI systems, to maintain gender, religious, and political hierarchies, are used. Fourth, it provides depth in history and structure, helping to frame Iranian AI practices within a broader history of militarized forms of government, religious authority, and global authoritarian learning. Finally, critical theory effectively informs this dissertation's qualitative and empirical methodology, particularly in relation to informed interviews with experts and activists, in the form of reflexive and context-sensitive knowledge production. Together, these reasons make critical theory not just a theoretical lens but an active framework for questioning and resisting the political rationalities embedded in AI systems.



4. Methodology

4.1 Qualitative Method

A qualitative data collection structure was chosen over a quantitative one as it allows for a deeper understanding of the events researched and how the study participants made sense of those events. To align with this goal, semi-structured interviews were conducted, as they allow flexibility and enable follow-up questions that offer deeper insights into participant experiences.

According to Sarantakos (1998), a qualitative case study allows for generalizations, as the findings may apply to the sampled population. Qualitative methods focus on understanding the 'how' and 'why' behind phenomena, rather than simply measuring variables (Sarantakos, 1998). This approach is especially relevant in politically sensitive environments, where understanding personal narratives and perceptions can uncover insights that structured surveys might miss.

To explore the relationship between AI and activism in authoritarian regimes, the research employed a qualitative approach. This provided a framework for investigating the impact of AI technologies from the perspectives of both AI experts and affected activists, rather than relying solely on statistical data.

Furthermore, my approach is from a dual position as a former political prisoner and an academic. I have been deeply involved in activism in the feminist movement and independent discourse. As such, I can bring insights and truths often marginalized in academic research. This session provides an opportunity to study how AI technology is felt, feared, and rejected.

4.2 Why Iran: The Condition of Freedom of Speech in Iran

The authoritarian regime in Iran has invested heavily in digital surveillance to identify and silence voices of dissent, and key institutions have emerged to develop new legal precedents, new technologies, and new detection mechanisms to capture cyber activists (Ünver, 2024). Iranian authorities have



drawn lessons from other digital authoritarian regimes, notably China, and have enhanced their technological capabilities to tighten internet control and construct a virtual firewall around users in Iran (Ünver, 2024).

Although dissenting voices persist, the regime has escalated its efforts to dominate cyberspace, with the ongoing crackdown on Virtual Private Networks (VPNs) representing the latest phase in this digital campaign for control (Lawfare, 2024). These cyberspace policies underscore the regime's recognition of digital control as essential to the preservation and endurance of authoritarian rule (Lawfare, 2024). This restriction is also applied to Iranian applications, including messengers and social media platforms (which belong to the private sector) (Ardebili Pour, Ghiasi, & Karkehabadi, 2024).

From the 2009 Green Movement to the 2022 'Woman, Life, Freedom' movement, Iranian users have used the internet, particularly social media, to raise their voices against the ruling autocracy (Tohidi, 2023). The Iranian regime's efforts at exercising control have encountered significant resistance. Iranian users have increasingly sought to circumvent censorship, which has prompted the government to develop a sprawling surveillance system to address the shortcomings of censorship measures (Tohidi, 2023).

As mentioned, I have first-hand experience with Authoritarian Repression. I have been repeatedly arrested and imprisoned in Iran for peaceful activism. This experience gave me deep insight into how authoritarian regimes operate and a nuanced understanding of how surveillance, censorship, and repression affect activists, including how technology is used in these processes. Such first-hand knowledge is invaluable in researching AI as a tool of authoritarian control, a phenomenon increasingly discussed in political science and AI ethics.

Iranian exiles and diaspora organizations are prominent targets; espionage and malware spear phishing are part of a broader arsenal of repressive tactics directed against human rights defenders and journalists outside the country



(Tohidi, 2023). Digital approaches are often combined with more traditional measures, such as pressure on relatives in the country and slander in state media (Ziccardi, 2013). State authorities aim to disrupt the connections of exiled activists with those inside the country and silence their ability to share information that challenges official narratives, both domestically and internationally (Ziccardi, 2013). As a result, they focus their efforts on prominent voices within the diaspora, particularly journalists affiliated with foreign Persian-language media outlets such as the BBC and Voice of America (VoA) (Michaelson, 2016). Central to these strategies of transnational repression is surveillance, which enables state actors to identify targets and tailor their actions based on the perceived level of threat.

4.3 Sources and Data Collection

The study collected primary data partly based on semi-structured in-depth interviews with relevant participants, including tech experts, activists, official reports, and policy documents from Iran on institutional reforms, policy proposals, and state-market relations. Secondary data was collected from national and international data that have already been published. Data is drawn from Western and Iranian media reports and existing literature.

• primary data

This study involved ten interviews, five with AI experts and five with activists, that in the next section, I have presented my desired definition of these two. As a data collection method, the same questions were asked of all AI experts and the same questions of activists. The interviews were carried out with stakeholders to gain insight into the context of Iran. To increase the likelihood of this attempt, the representatives were chosen from different genders, minorities, and religions. The reason for this method is that it gives a broader perspective on the situation from various angles.

Some interviews were conducted online via Google Meet, with follow-up questions. The semi-structured interview technique has been used since it is suited for a bottom-up approach to collecting information. This study uses



semi-structured interviews, triangulation, learning from the stakeholders, and a review of secondary sources. The excellent knowledge of democracy, the political situation, and the development of AI in Iran among the interviewees effectively minimized the risk of bias. Since the study aimed to collect responses that, despite political orientation, were not based on personal bias but rather on the impact on freedom of expression

- AI expert

"A professional with advanced knowledge and experience in designing, developing, and applying artificial intelligence systems and techniques, such as machine learning, natural language processing, computer vision, and robotics. AI experts often hold advanced degrees in computer science or related fields and work in research, academia, or industry."(European Commission 2021)

This research draws on interviews with five IT experts, including a Senior Researcher in Ethical AI and Machine Learning Interpretability, the Chair of Artificial Intelligence Policy and Governance, a Lead Scientist in Deep Learning and Computer Vision, the Director of AI for Development and Inclusion, and an Associate Professor specializing in Algorithmic Bias and Fairness in AI Systems. All interviewees began their academic journeys with undergraduate degrees in fields such as computer engineering, computer science, or electrical engineering. Due to technological and governmental constraints on technology-related disciplines in Iran, they pursued their postgraduate education abroad. Although currently living in exile, these experts remain deeply informed about Iran's socio-political and technological landscape. Their professional work spans various institutions and organizations that focus on challenges in developing countries, and they have actively contributed to initiatives aimed at reducing state pressure on activists.



- **Activist:**

“An individual who takes deliberate action to promote, impede, or direct social, political, economic, or environmental change. Activists may act individually or as part of a group, using various forms of protest, advocacy, or public engagement to influence public opinion or policy.”(Oxford Reference 2023).I define activism by modifying Yannis Theocharis's (2015, 6) term to cover citizens' voluntary actions to raise awareness about or exert pressure to solve a political, cultural, or social problem. Our definition thus covers various issues, from social and environmental problems to human rights, local disputes, and more.

In this research, we interviewed five activists and tried to have them come from different backgrounds. They are Women's Rights Activist and campaigners against Compulsory Hijab, Labor Rights Advocates and Spokesperson for the Unofficial Workers' Syndicate, Kurdish Language and Cultural Rights Educators and Political Prisoner, Environmental Activist Opposing Water Mismanagement in Khuzestan, Student Activist and Organizer of University-Based Democracy Forums. All of them have been repeatedly persecuted and detained and are aware of the new types of security pressures that have increased with technology. They all live in Iran, and due to internet restrictions, interviewing them took a long time, but their first-hand and valuable information was invaluable for this thesis.

The interviews have been based on key and open-ended questions to leave room for conversation, although still being controlled and structured, and to allow the interviewee to feel as free as possible to express their concerns and feelings. The interview procedure usually started with a more informal discussion about who we are and our reason for the visit, followed by the purpose of the study, and then moved towards a more formal interview.

Triangulation is used between our interview results, previous research from various sources and disciplines, and our reflections to increase the validity and reliability and to put our findings in a broader context. To increase the



validity and reliability of the study, a literature study with secondary sources was conducted before and during the study, and to double-check all facts, a more holistic and cross-disciplinary perspective was obtained.

Transparency with all interviewees was pertinent for the study to achieve its objective and receive adequate responses. The privacy considerations of the interview tool were critically examined to ensure the confidentiality of the data/answers. Lastly, the respondents received an email with the study's objective and ethical considerations before the interview, which was significant for consent.

The duration of the interview was 40 to 60 minutes, and the respondents selected a suitable date and time. This study contains audio media; the transcriptions of the interviews were in text form. The language used for the interviews was Persian. Since the Persian language was used for interviews, recording and transcribing all interviews and sending them back to the respondents for review was essential. According to the six-staged procedure developed by Braun & Clarke (2006), interview transcripts were examined through thematic analysis.

● **Secondary Data**

In order to broaden the empirical basis beyond the interviews, thesis rests on an extensive number of secondary sources, such as international human rights reports, policy briefs, academic works, and noteworthy media reports. These sources round out the discovery and point to a wider picture of Iran's AI-driven tactics for repression.

Critical Reflection on Secondary Data As with any data, secondary data is not neutral. Although these secondary sources provide crucial information on Iran's AI community and repressive tactics, they are not without their own limitations. Much of the coverage is based on witness accounts without official confirmation, or is written under a security clampdown preventing independent verification. Moreover, most international models are predicated on democratic institutions and oversight, which do not exist in



authoritarian states such as Iran. Secondary data therefore cannot do more than give us a partially embodied sense of what it is like to be subject to these technologies.

4.4 Thematic analysis

Thematic analysis is a widely used method in qualitative research for identifying, analyzing, and interpreting patterns of meaning within data (Nowell et al., 2017).

According to Braun and Clarke (2013), thematic analysis offers a flexible and accessible method for analyzing qualitative data. This method is particularly valuable for novice researchers seeking structured guidance in handling interview material. We use thematic analysis When our research question seeks to understand experiences, perspectives, or meanings. We use Code inclusively, meaning we capture anything that might be relevant. On the other hand, themes are not just topics; they should say something meaningful about the data and research questions, and we should be aware of our influence as researchers.

The first step, thematic analysis, includes transcribing interviews, reading and re-reading the data, and making initial notes. Then, systematically code interesting features across the dataset and Collate codes into potential themes related to our research question. After that, Check that the themes work with the coded data. Finally, Combine analytic narrative and data extracts to tell a coherent story about the data.

4.5 Interpreting AI and Activist Perspectives Through Thematic Analysis

Thematic analysis (TA) is used to analyze text data collected from semi-structured interviews with AI practitioners and advocates. TA is particularly suitable for this thesis because it converges to computer science, human rights, and political activism. There is the advantage of its pliability in addressing both technical knowledge and lived experience, as specialists from various fields contributed to topics such as predictive analytics, face



recognition, and natural language processing, especially as applied to state surveillance. Instead, activists recounted their own experiences of digital repression, such as being censored, the internet being shut down, and being watched with unsecured phones and messaging. The TA facilitated an analysis of the rich themes in both types of knowledge while retaining the complexity of each view.

In addition to detecting patterns, TA made it possible to relate individual stories to the larger socio-political processes. It enabled the identification of overarching themes of algorithmic discrimination, digital disobedience, and the instrumentalization of surveillance that ran through expert and activist testimonies. It also exposed forms of resistance under authoritarian circumstances and demonstrated that AI infrastructures are not only instruments of repression but also spaces of opposition.

The analysis was sensitive to intersectional conditions of vulnerability, such as being subject to gendered surveillance and linguistic discrimination against Kurdish activists. Slight but complex risks, such as these, could easily be hidden in broad groupings, and TA helped surface them.

Importantly, TA aligns with the critical theory framework of the thesis, which examines power and domination. In this light, TA does more than aggregate information. It highlights how AI technologies are implicated in authoritarian political processes, how they can algorithmically silence dissent, deepen systemic inequality, and, paradoxically, also give rise to novel forms of resistance. As such, TA in this research does double duty, acting not only as a methodological tool but also as a critical lens through which to analyze AI's political work in repressive contexts.

4.6 Generating Initial Codes, Coding Insights from AI Experts and Activists

To operationalize the thematic analysis, the data were systematically coded from two groups AI experts and activists, each offering unique but



complementary insights into the deployment and contestation of AI systems in Iran.

For the qualitative data from the semi-structured interviews, a common coding framework was established, drawing on insights from both AI experts and civil society activists. This structure enabled the exploration of common themes and the specific positionality and concerns of each group. In the process of coding, codes were derived inductively and gathered around overlapping categories, including the implementation of technology, repressive strategies, resistance practices, and sociopolitical implications.

Interviews with AI experts were transcribed, capturing the technical basis of surveillance and algorithmic governance. These involved AI-based facial and crowd surveillance, such as using computer vision and gait recognition, as well as AI-driven monitoring for online dissent via natural language processing (NLP) and proactive risk profiling through predictive analytics. They also highlighted more advanced methods, such as social graph analysis, to map activist networks and infrastructure deployments, including facial recognition on public transit systems. Thoughtful posts about platform bias, the misclassification of protest-related content, and the absence of legal protections in Iran further contextualize how AI tools mirror and entrench historical power imbalances. There were also concerns about the manipulation of generative AI for disinformation and harassment, as well as mobile surveillance, including spyware and packet inspection.

The activist interviews, in turn, offered a bottom-up view of what it means to live with and struggle against these technologies. [Participants mentioned how] gendered AI surveillance was used in tracking hijab in public space. However, they also detailed the politicking of AI content moderation, which squelched hashtags or shut down accounts at politically sensitive times, including instances where the technology switched offline hashtags during the Mahsa Amini protests. Tactical resistance tactics, such as the use of burner phones, encrypted messaging apps, Faraday bags, and decentralized organizing, were among those cited by activists and organizers most often.



Other codes pointed to the growth of state repression via automated monitoring and SMS warnings, ethnolinguistic biases affecting Kurdish content, and biometric surveillance of environmental activists. In some places, such as university campuses, locations were chosen for targeted repression through actions such as the use of facial recognition **Closed-Circuit Television (CCTV)** and state-controlled blackouts of the Internet to hinder mobilization. Importantly, interviewees highlighted grassroots digital security efforts, including peer-to-peer education, encryption, and circumvention. And pointed to how legislative instruments, such as the User Protection Bill, became tools for legitimizing and institutionalizing digital authoritarianism.

These codes collectively served as the basis for the thematic analysis, which in turn led to the emergence of higher-level themes, including AI as a shortcut to facilitate authoritarianism, algorithmic discrimination, resistance and resilience in digital spaces, and the legal codification of surveillance practices. This combined framework reflects the multidimensionality of AI and provides a lens through which to view not only how technological apparatuses and social facts intersect in Iran, but also how power structures ultimately shape these relations.

4.7 Defining and Naming Themes

In accordance with Braun and Clarke's (2006) thematic analysis approach, this section defines the latent themes derived from the coding phase. Each theme encapsulates a particular dimension of AI's role in Iran's surveillance state and civic resistance.

The findings of a thematic analysis of qualitative interviews with Iranian AI activists and AI experts are discussed in this chapter. Its analysis uncovers six interlinked themes serving to illustrate how the Iranian state is operationalizing artificial intelligence technologies to create a surveillance environment in which the state's opponents are monitored, censored, and silenced. At the same time, activists seek to resist through digital and offline



strategies. Descriptive Themes Each theme is described and illustrated with verbatim data extracts, and interpreted about the research questions of this study.

4.8 Ethical Considerations

“I think it very important that the mere fact of there being surveillance takes away liberty. The response of those who are worried about surveillance has so far been too much couched, it seems to me, in terms of the violation of the right to privacy. Of course, it’s true that my privacy has been violated if someone is reading my emails without my knowledge. But my point is that my liberty is also being violated, and not merely by the fact that someone is reading my emails but also by the fact that someone has the power to do so should they choose. (...) It’s no use for those who have possession of this power promising that they won’t necessarily use it or will use it only for the common good. What is offensive to liberty is the very existence of such arbitrary power”. (Skinner, 2013 cited in Bernal, 2016, p.250)

The Puerto Rico Conference in January 2015 was a landmark event to promote AI's beneficial and safe development. It led to an Open Letter signed by over 8,000 people calling for the safe and beneficial development of AI and a research agenda to that end (Future of Life Institute, 2015a). The Asilomar Conference in January 2017 led to several thousand AI researchers signing the Asilomar AI Principles (Future of Life Institute, 2017a).

Traditionally, ethics is defined as the “code of moral principles that sets standards as to what is good or bad, or right or wrong in one’s conduct and thereby guides the behavior of a person or group” (Schermerhorn, 2002, p. 146). Researching activists involves ethical considerations. The highest priority was a cautiously planned method, and necessary steps were taken to ensure the safety and integrity of the study participants and the data provided.

After the data were interoperated, the interviews were conducted ethically to protect the respondents and provide more transparent answers. It is a moral



standard to consider that some respondents may have political or personal reflections in their answers. The privacy and ethical concerns of the participants were taken seriously. A guideline was used to inform the respondents of the motive of the research and its objective to increase the study's legitimacy. This way, the respondents could participate in the study based on their time and effort in answering the questions. The respondents had the right to see all the questions before data collection and decide whether to participate in the research.

While some respondents may want to be recognized for their contributions, others may wish to remain anonymous. Therefore, there is an obligation to keep the respondents' identities safe and secret. Because of the limitations of democracy and freedom of speech in Iran and because this study was oriented to AI and activism, the anonymity option of respondents was necessary to hinder anyone from tracing their identity.

In all cases, the respondents were informed while recording their responses. The purpose of the recording was made known, and the respondents were assured that it would only be for transcription and translation. The recording was used to double-check the answers and send them back to the respondents to review and change/remove anything they felt uncomfortable with.

Ethics were critical, given the sensitivity of the work in Authoritarian atmosphere research on state repression and surveillance in an authoritarian environment.

The study complies with the ethical principles of voluntary participation, informed consent, and confidentiality. All subjects were given an informed consent form explaining the purpose of the study, the potential risks involved, their rights, and the option to withdraw from the study at any time without penalty.

To ensure the safety of participants, particularly Iranian activists and experts, no names or identifying details were documented or published. Interviews were de-identified, and participants could refuse transcription and



instead opt for field notes to be taken. Given the sensitive nature and potential emotional distress associated with recalling accounts of digital repression, participants were explicitly also informed that it was up to them to skip any question.

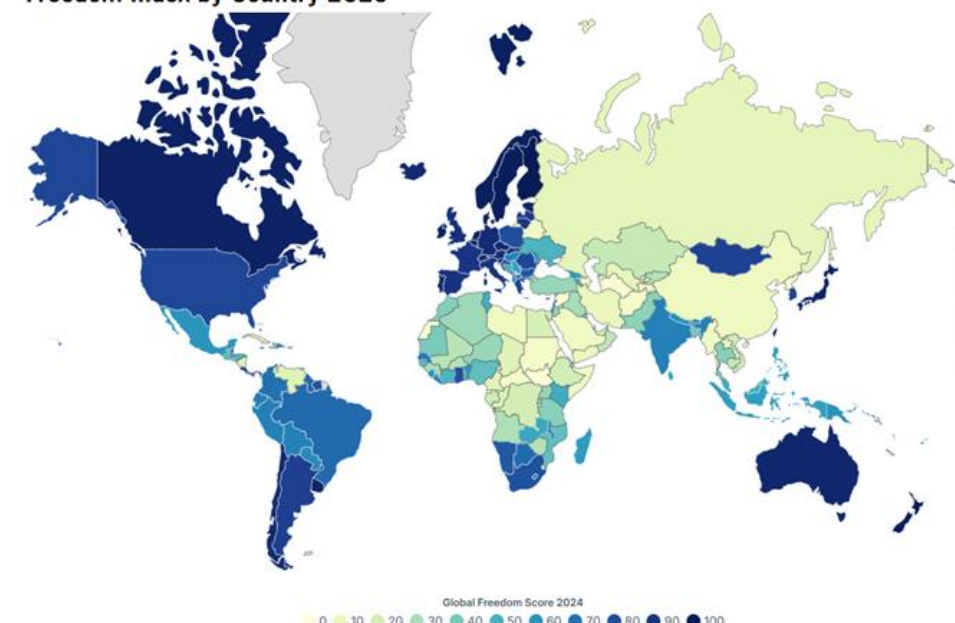
Taking these steps would ensure that the research process did not violate the right of participants to self-determination, as well as their safety and integrity (Linnæus University, 2010), as recommended in both Linnæus University's research ethical policy and human rights research internationally.

4.9 Limitations and Delimitations

Limitation

Iran is the eleventh country with the lowest freedom of expression and information score according to World Population Review, 2025. This posed significant limitations on data access, trust-building, and participant recruitment.

Freedom Index by Country 2025



Source: World Population Review. (2025)



Iran's national public service delivers news upon loyalty to the government; therefore, a lack of independent news media is prevalent. However, all sources were critically examined and used cautiously to counter this limitation.

Most AI surveillance programs and repression strategies used by authoritarian states are not publicly documented or classified. This restricts firsthand system design, deployment, and state intent data visibility. Although this may limit nuance or context, it is necessary for policy documents or media content published in languages other than English (e.g., Farsi).

The stories of activists or government insiders in authoritarian regimes come with immense ethical and safety concerns when conducting interviews in the field. Therefore, this option will not be available or will be significantly limited. NGO reports and journalistic sources are vital, but they are not always verifiable, often due to security reasons. This makes triangulating findings a challenge.

Another limitation is the biased effect on respondents' answers based on their social status; for instance, all the activists we interviewed were against the Islamic regime, which could affect their answers. The greatest challenge has been conducting the field study since our inexperience in interviews made it challenging to be fully aware of bias and avoid common mistakes. Mikkelsen (2005: 176) explains common mistakes as a failure to judge answers, repeating and asking vague questions. The situation when the interviews were conducted has to be considered since the situation can have influenced how the answers and our questions were presented. Concrete problems such as internet problems in Iran sometimes interrupted interviews may have influenced the interviews.

Delimitation

The research focuses on Iran as an authoritarian regime using AI technologies to govern and control. While perhaps suggestive of global



authoritarian trends, the data do not pretend to be able to generalize to worldly phenomena unless explicitly comparative. Also, we are confined to narrow AI uses, specifically facial recognition, predictive policing, and AI-driven content moderation. More general or hypothetical considerations of artificial general intelligence (AGI), superintelligence, or military AI systems fall outside the scope of this work.

The study is limited to authoritarian and hybrid regimes with weak civil society and focuses on non-democratic uses of AI. It deliberately steers clear of considering nations' regulations and ethics in fully democratic systems. It uses critical theory and Science and Technology Studies (STS). It does not directly challenge narrow technical evaluations of AI competence or algorithmic precision or critique purely commercial drives to implement AI. Interviews are held with AI experts and activists but not state officials, law enforcement, or private AI companies directly involved in government repression due to access and security risks. Our focus on 2015 and after, since that year, the inclusion of AI in governance at international and domestic levels in Iran has begun to rise. For earlier history, readers should refer to the relevant article where the concepts are presented in a historical context. Interviews and internal documents in Persian predominated, reporting on Iranian socio-political relations. For this reason, the study does not include voices from Iran's diaspora unless they advocate for something directly taking place inside the country. The study is qualitative (primarily semi-structured interviews). It does not involve extensive surveys, social media content analysis, or computational modeling, though we feel those subjects may be relevant for future work.

5. Finding

“There is no such thing as privacy on the Internet anymore” (Merkel, 2014).

In this section, we present our findings, obtained through 10 interviews and an in-depth review of key documents. These findings illuminate the multi-



layered interplay between AI technologies and authoritarian governance, particularly in the context of the Islamic Republic of Iran.

5.1 key findings from interviews

5.1.1 AI Expert

Interviews with AI experts identified several key themes that align with the research questions (RQs) and theoretical framework. These themes illustrate how authoritarian regimes, such as the Islamic Republic of Iran, use AI technologies not only for monitoring but also for coercive control of civil society.

One AI expert reflected that: “The tools used to develop and achieve this capability are diverse.” All agreed that authoritarian regimes are pursuing multiple technical avenues to enhance their surveillance capabilities. These include facial and gait recognition, behavioral analytics, natural language processing (NLP), predictive policing, and social network analysis. (e.g., hijab enforcement in metro and traffic systems). Artificial Intelligence Used for Surveillance and Control and Natural Language Processing (NLP) for analyzing speech, sentiment, and online dissent in real-time, including Persian content. Another expert mentioned, "Preemptive targeting of protest hotspots to identify individuals using predictive analytics and graph learning and social network analysis to extract relationships between activists and political opponents has been used repeatedly in recent protests, and because it is time-consuming, it has led to some people being arrested weeks after the protests."

Biometric identity systems, such as those in Iran (Sajam, Amayesh), are another way to limit the presence of activists through behavioral surveillance. These are preventive capacities, not merely reactive ones of authoritarianism. Artificial intelligence allows them to act before dissent becomes public. Expert 2 believed that Iran is “a browser market” for global digital authoritarian trends but also adapts existing technologies to its



domestic context. Iran often imports or copies Chinese and Russian technologies but is increasingly developing domestic alternatives to reduce its dependence on foreign aid. These practices reflect Iran's efforts to integrate imported knowledge with domestic innovation, creating an independent surveillance ecosystem.

Experts also noted that smart city technology in Tehran paves the way for AI surveillance as part of a modernization program. AI-based surveillance tools are embedded in the local information network that connects Iranian host companies with global services. Facial recognition enforces ethics laws and tracks protesters in the Surveillance Plan (Scheme Noser) and traffic monitoring systems.

Three of the interviewees emphasized that AI systems in Iran can not be technically neutral. Its deployment within authoritarian settings transforms it into a tool of repression. Expert 3 said, "Even when using GPT chat in Iran and other non-free countries, it can still contain misinformation. In Iran, only state media is allowed to operate, and most information about the situation in Iran, whether in Persian or English, is produced by the government. When you use OpenAI, it responds based on the data and content available, and as a result, many of the explanations and answers may be biased."

Key risks include data bias that disproportionately targets minorities, the misclassification of activism as extremism, and the branding of dissenting speech as "inauthentic behavior." These findings corroborate the argument that algorithmic systems in oppressive contexts amplify structural. Expert 2 highlighted: "We as engineers were designing algorithms to optimize, not suppress. But I saw how the same technology was put at the disposal of security agencies to monitor activists. This is a misuse of science." Another theme in the interviews was the failure of international ethical AI frameworks to counter authoritarian abuses. Expert 3 believes that current standards (e.g., Human-Centered eXplainable AI (HCXAI), Organisation for Economic Co-operation and Development (OECD), and Institute of Electrical and Electronics Engineers (IEEE)) assume a form of democratic



oversight unavailable in Iran. Furthermore, voluntary guidelines lack authority and can be easily circumvented or manipulated by authoritarian governments. Expert 2 added that Iran has yet to enact domestic data protection laws, and the passage of laws such as the UPCOS bill increasingly codifies control under the guise of user protection. This further demonstrates that ethical AI is an ineffective tool for reducing digital repression without global accountability.

Expert 4 reported: "Iran is using artificial intelligence not for development, but for control. What is used for mass surveillance in China has become a tool for individual silence in Iran." another expert emphasized," Due to the need for significant investment, governments worldwide are the primary consumers of technology, and the Islamic Republic has imposed additional restrictions on AI education in academia. Although acceptance of a doctorate usually requires approval from security agencies, advanced AI education in Iran, such as nuclear physics, is also subject to additional restrictions." While the stakes were high, experts also saw early evidence of resistance and adaptation from activists and technologies. For example, Expert 3 stated that adversarial machine-learning tools are being used to disrupt facial recognition algorithms. Activists can still communicate securely on platforms like Signal and Briar. Digital literacy campaigns run by expatriates abroad also allow activists to understand and manage surveillance risks. But they all agree that such efforts are gradual and under-resourced, especially given the efforts of state actors to wage war in cyberspace. However, expert 4 warned that AI advancements, such as deepfakes and biometric-linked public services, could intensify state control. Combined with cross-platform censorship, these tools may entrench AI in everyday governance. The convergence of generative AI and biometric surveillance signals a future of totalizing repression, necessitating enforceable global norms.

5.1.2 Activist

Interviews with five Iranian activists from women's rights, labor, environmental, cultural, and student movements revealed that each



experienced AI-enhanced repression in overlapping yet distinct ways. The findings provide an empirical and rooted understanding of how repressive AI infrastructures are experienced by targeted populations in Iran.

One of the activists started his interview with this sentence: “The threat of AI exists everywhere in the world, but journalism associations, human rights groups, and activists are constantly reaching out to governments, protesting and criticizing them to prevent governments from misusing AI. However, the main problem in Iran is the lack of a free civil society and the restrictions on journalists and activists who cannot protest due to repression. Even universities and the academic environment are in the hands of the government, and academic freedom does not exist; therefore, academic experts cannot speak clearly about the dangers of AI to freedom.” I believe it was the core of all activists' interviews. All of the activists interviewed described encounters with surveillance technologies that are increasingly AI-based. They noted facial recognition, license plate readers, and drone surveillance being deployed during protests and public demonstrations, reports of women being identified by traffic cameras for violating the hijab rule and receiving automated text messages.

Labor activists also faced the threat of geo-tracking through mobile phone apps and text messages after strikes. Kurdish and student activists pointed to the use of AI-based surveillance in dormitories, schools, and educational platforms, which have been created to shape a culture of constant control. These reports confirm academic claims that “smart cities” are being repurposed as surveillance architectures in authoritarian environments (Akbari, 2022). Activists denounced content moderation as a tool to suppress free speech, arguing that it was being implemented through domestic controls and algorithms on international platforms.

One of the activists noted that “Instagram, Telegram, and Twitter posts about protests or cultural rights were removed, blocked, or blocked. Machine learning moderation systems, designed without considering the nuances of Persian or Kurdish, often mistakenly labeled activists’ content as extremist or



disinformation.” other activists, referring to Carol Hanisch, argued, “the personal is political” (Hanisch, 1969), highlighting the political significance of personal experiences. She emphasizes: “In addition to social activities, we also do not feel safe in our daily lives because the security system can also exploit our private issues.” Meanwhile, pro-regime disinformation campaigns were not addressed, suggesting that moderation rules were unevenly enforced. This chills political discourse and undermines trust in international technology platforms, proving that even well-intentioned algorithms can perpetuate authoritarian narratives.

Despite the constraints, activists across sectors demonstrated digital resilience by adapting their strategies. Two activists mentioned they have used standard tools, including VPNs, encrypted messengers (e.g., Signal, Briar), Burner phones, and Faraday bags. Another activist also emphasized that offline methods such as face-to-face organizing, coded communication, and physical leaflets remain vital—especially during internet outages.

In some areas, activists trained their peers in digital literacy, metadata awareness, and operational security. However, access to secure tools is limited due to sanctions, costs, and digital illiteracy. There is a pressing need for local, open-source tools and resources in Persian or Kurdish. Another activist believes, “ When there is no freedom of speech in a country, the government's ability to misuse technology is even greater. As Marcuse said, “The small, powerless minorities who struggle against the prevailing ideology and its institutions must be helped: their continued existence is more important than the preservation of abused rights and liberties which grant constitutional powers to those who oppress these minorities.”(Marcuse, 1969, p. 109)

The student activist warned: “In Iran, your face is captured by city cameras, your voice is recorded from Instagram stories, and before you know it, you're expelled from university. Artificial intelligence is the long arm of repression.” Activists emphasized that Iran’s legal and infrastructural frameworks are engineered to facilitate digital repression intentionally. The



Consumer Protection and Core Online Services (UPCOS) bill legitimizes government requests for surveillance while requiring platforms to comply with national censorship requirements (Article 19, 2022). The National Intelligence Network is a homogeneous structure that isolates Iranian users from secure global networks and facilitates state ownership of telecommunications providers, universities, and municipal services, as well as cross-sectoral data aggregation and population profiling. In addition, one of the activists describes her feelings in this way: "When you film violence against women with your mobile camera, you know that a facial recognition system may identify you at several stages. Artificial intelligence is not a tool for us, but a threat to the very nature of seeing and speaking."

The institutionalization of surveillance transforms everyday services into instruments of repression. Interviews with Kurdish activists demonstrated the differential impact of AI surveillance on activists, with ethnic minorities (e.g., Kurds, Arabs) and gender (women) often being perceived to be more suspicious or noticed. Kurdish language content is heavily censored, and surveillance has increased in border regions such as Kurdistan and Khuzestan. Environmental and student activists described being considered "threat agents" and having digital evidence used to arrest or deport them. This confirms Critical Theory's focus on the power asymmetries embedded in technology systems, especially where these technology systems are uncontested. Activists I interviewed articulated some everyday demands. They require technical support to develop tools for local, censorship-resistant user interfaces with privacy by design.

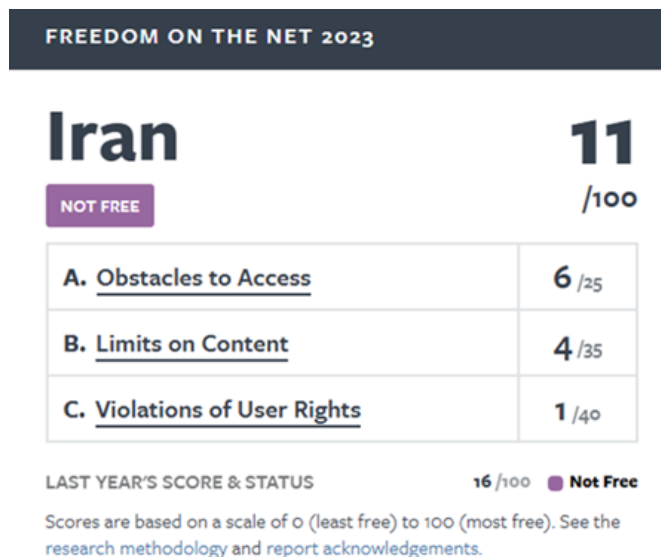
Additionally, they should repeal repressive laws like the Consumer Protection and Core Online Services Bill (UPCOS) and advocate for data minimization and transparency in all surveillance systems. Forcing international tech companies to audit their moderation algorithms, reducing collaboration with authoritarian governments, and helping fund robust infrastructure for at-risk users are other requirements. Ultimately, this would mobilize activists, human rights groups, labor unions, environmental



coalitions, and academic organizations to support the inclusion of digital rights in broader advocacy efforts.

5.2 key findings from documents

Iran is a case in point, in which the architecture of digital authoritarianism is already well-entrenched. Iran is listed among the worst countries on internet freedom by Freedom House (2023). Their report details widespread censorship, invasive surveillance, content manipulation, and user harassment elements that sound all too familiar from Iranian activists' testimony. Crucially, it says there has been a significant worsening of online freedoms after the 2022 unrest a move towards a level of repression similar to China's use of AI to silence critics.



source: Freedom House (2023)

They are deeply gendered applications of this authoritarian use of technology. Amnesty International (2024) and the UN Fact-Finding Mission (2025) have reported on the use of facial recognition and other AI-based surveillance to enforce Iran's mandatory hijab laws. These results are consistent with interview data indicating that AI is deployed to police women's appearance and limit their mobility. Institutionalizing this gendered



mode of control are tools like surveillance drones, public cameras, and citizen-reporting apps such as Nazer.

Academic studies and papers from organizations like the V-Dem Institute and Insikt Group (2025) elaborate more deeply on how AI helps to reinforce authoritarian control through means like predictive policing, facial recognition, and social graph analysis. These tools enable regimes to identify and stamp out dissent before it coalesces. These mechanisms, according to Frantz, Kendall-Taylor, and Wright (2020), not only help suppress individual resistance but also serve to strengthen the structural dominance of their rule over political elites and the opposition at the grassroots level.

Although Iran does not have a distinct ethical or legal framework for AI governance, it has already been involved in international discussions on AI ethics. Organizations such as the Future of Life Institute (FLI) have emphasized the risks of exporting AI technologies to authoritarian states without adequate safeguards. The Asilomar AI Principles (Asilomar AI Principles, 2017) promote value alignment and warn against the unrestricted deployment of AI in repressive situations. These worries lend credence to the argument of this thesis that AI is not a neutral implement but a political technology entwined with hegemonic power relations.

Independent media, including BBC Persian, IranWire, and Al Jazeera, among others, offered indispensable contemporaneous documentation of AI-enabling state abuses, including internet shutdowns, hijab-linked surveillance, and activist detentions. Here are reports that close the gap between government policy and the human reality they spawn, especially when, as following Mahsa Amini's death and the surge in surveillance, they come home.

The Supreme Council of Cyberspace (SCC), established in 2012 under the leadership of Supreme Leader Ali Khamenei, has played a key role in normalizing digital repression. This council comprises the highest-ranking



officials of the government, representatives of the Islamic Revolutionary Guard Corps (IRGC), members of the judiciary, and ministers of defense and information and communication technology (ICT). Regulating digital governance in Iran, the SCC wants to develop a closed “National Information Network (NIN)” that effectively censors Iran from the internet. The ‘Woman, Life, Freedom’ movement of 2022 ratcheted up the regime’s stake in such repressive digital edifices.

Freedom of speech is theoretically guaranteed by the Iranian constitution, according to Article 24, but is limited in practice. Vague legal clauses restrict the criticism of Islamic tenets or the rights of citizens, facilitating widespread censorship (Penn Today, 2012). Other repressive legal instruments against speech are:

The Press Law (1986): Targets material defined as divisive or harmful to the Islamic Republic.

The Islamic Penal Code: Criminalizes Vaguely Defined “propaganda against the state.”

Computer Crimes Law (2009): Prohibits the distribution of online content under ambiguous criteria such as “public morality” (Islamic Republic of Iran, 1979; 2009).

The laws are implemented by various state authorities, including the Ministry of Culture and Islamic Guidance, the judiciary, and the IRGC (Gulf Centre for Human Rights & International Human Rights Law Clinic, 2021).

Iran is eagerly developing domestic surveillance based on Chinese designs. It describes a ‘centralized monitoring system’ that collects lifestyle data by making access to databases mandatory (Ensaf News, 2023). The Cyber Police (FATA) is one of the main repression machines that, in 2011, was responsible for identifying and collecting data to arrest demonstrators by using online spying and monitoring software (Mashregh News, 2019; Tasnim News, 2019). Budget from online transactions (Digiato, 2022) support its capability.



The domestic development of AI infrastructure has accelerated. Spyware named EyeSpy, found inside 20Speed and other such pirated VPN apps, was exposed in January 2023 by the cybersecurity firm Bitdefender. This malware steals documents, images, and even crypto-wallets and with it, severe privacy breaches. In 2024, Iran announced the establishment of its National Artificial Intelligence Organization, accompanied by significant funding and 40 policy documents. In 2025, a prototype of a national AI platform was introduced, designed to be localized in partnership with Sharif University (Iran International, 2025).

AI-enabled weapons and war machines are integral to law enforcement initiatives for governing remote surveillance and decentralized governance infrastructures (Lawfare, 2024; Mieses, Kerr, and Jahanbani, 2024). Yet, indigenous counterfeited VPNs – labeled by several insiders as a “Mafia of VPNs” – contribute to both user information theft and cyber entrapment (Didban Iran, 2021).

The Iranian model is a global example of how digital authoritarianism spreads. Advanced surveillance systems permit regimes to watch internet use, gather sensitive personal data and follow dissidents at home and abroad. They are also evidence of how today, authoritarian power extends across borders to suppress dissidents and journalists living in exile around the world.

6. Analysing

6.1 Patterns of Digital Repression and Digital Resistance

Below, themes reflect the perspectives of both AI experts and activists while illuminating how digital authoritarianism in Iran is the product of both state-engineered technological domination and bottom-up struggles. Interpreted with the lens of Critical Theory, these practices demonstrate that AI is not simply a benign technology but an extension of hegemonic power used to surveil, classify, and discipline political life as it generates a new subjectivity and resistance as well.



Theme 1: Surveillance Technologies as Power

Among the key technologies the expert observed the Iranian regime utilizing to carry out surveillance were computer vision, natural language processing (NLP), social graph analysis, and predictive analytics. Although these instruments reflect what is more broadly a global trend in digital authoritarianism, their use in Iran is especially opaque and unregulated. Methods developed for urban control, such as creating “smart” traffic systems, have been applied to political monitoring. Tools like facial recognition cameras in metro stations and license plate scanners on highways provide real-time insight into the movements of the city’s inhabitants. Analysts also discussed how predictive analytics aren’t used reactively but proactively, taking a citizen's profile and anticipating dissatisfaction before it arises. These artworks exemplify Critical Theory’s insistence on preemptive regulation in contemporary power relations, wherein surveillance is inscribed into daily existence in the name of efficiency and order.

The technologies are not inherently political, but political priorities heavily influence their application in Iran. However, experts have emphasized that in a system in which dissent is deemed a national threat, AI systems are often built to flag activism as extremism. Systematic mislabeling arises from the very biased datasets and exclusionary design, which mainly means un-/under-representing groups (like Kurds and non-Persian speakers)—facial recognition algorithms, which are proven to be used to identify and arrest protestors. Western-style censorship systems also fail to understand the cultural and political nature of their activism, resulting in even more silence. This dynamic is a classic example of the phenomenon in Critical Theory known as the “false neutrality” of technology: systems that claim objectivity but reproduce and exacerbate the existing social hierarchies and political exclusion.

Theme 2: Legal Vacuums and Systematized Impunity



Lacking democratic oversight also drives abuses of AI, experts added. Whereas international entities promote ‘ethics in AI’ and ‘privacy-by-design,’ these models rest on the premise of working democratic institutions — not an ancillary part of Iran. Laws such as the User Protection Bill do not protect but rather legalize surveillance and censorship. "Rule of law without accountability guarantees is an empty shell, one that sanctifies mass surveillance and lets technologists off the hook." Experts stressed that international sanctions and binding global norms would be required to end this kind of abuse and underscored the structural character of repressive rule, a key theme in Critical Theory, which denies that power can be legitimate when exerted without either democratic consent or accountability.

Theme 3: Increasing Vulnerabilities, Resistance in the Making

Also on the horizon, experts cautioned that generative AI, smart city technologies and biometric platforms could be weaponized to facilitate deeper profiling, including disinformation campaigns and intimidation. However, they also identified emerging counterstrategies, including adversarial machine learning to foil facial recognition, poisoning datasets to confuse surveillance models, and diaspora-driven development of privacy tools. However, these innovations are piecemeal, under-resourced, and largely unsupported by the institutions. This tug-of-war is a reminder that a dialectic of power and resistance is at play, in which the tools of repression can ironically become the catalyst for subversive strategies and communities of care.

Theme 4: Banal Surveillance and the Criminalization of the Nitty-Gritty of Civic Life

In their recounting, activist accounts emphasized the pervasiveness of surveillance in everyday life. Traffic systems, public transit, and university facial-recognition cameras were listed as instruments of continuous surveillance. Women’s rights activist said she had received anonymous SMS warnings that appeared to be based on hijab-detection software, while



Kurdish activist described border surveillance systems to profile ethnic groups. These narratives are a testament to how biometric and geolocation technology is deployed to criminalize participation in public life, transforming things like acts of visibility , walking unveiled, joining a protest, into punishable offenses.” As Critical Theory has implied, the body is disciplined to become a subject controlled, in which technologies of surveillance attempt to standardize submissiveness and marginalize dissent. Activists are bothered by both official censorship and the automatic censorship of global platforms. Activists say that posts were consistently taken down, and protest hashtags and accounts were shadowbanned during the Mahsa Amini protests. AI moderation tools, trained on Western data and oblivious to regional context, often misclassify genuine activism as harmful or violent content. Together with Iran’s National Information Network (NIN), this has produced a fragmented and securitized Internet space in which free expression is increasingly stifled. These practices demonstrate how algorithmic governance functions as a transnational stratum of repression, extending state violence through effectively decontextualized corporate tools.

Theme5: Subjugation Resistance From Below and Digital Insurgency

Despite harsh suppression, Iranian activists still manage to create while squeezed. Respondents employed encrypted messaging applications (e.g., Signal, Briar), disabled biometric features of their telephone devices, and used multiple “burner” phones to avoid detection. They plan self-defense workshops and produce materials that are not dependent on the Internet, and they coordinate in a decentralized way. However, access to these resources is uneven and is even hindered in rural and border areas, where sanctions and poor connectivity further exacerbate the issue. These practices are an example of prefigurative politics or the creation of alternative social arrangements in the fissures of authoritarian domination. They also highlight how tenuous resistance efforts can be when faced with political and infrastructural limitations.



Taken together, the interviews describe a multifaceted ecosystem where AI technologies, far from being neutral tools, are deeply entwined in the reproduction of authoritarian power and the restriction of political agency. Using Critical Theory, we can see that these technologies not only discipline but also make visible the outlines of resistance, where willful acts of refusal, creativity, and solidarity manifest in the face of systemic domination.

6.2 Mechanisms of Digital Control and Strategies of Resistance

Based on findings from interviews with AI experts and civil society activists, the analysis is structured through two interrelated dimensions: the first is the enabling and deepening of state repression by AI technologies, and the second is the counter practices developed by activists to reclaim space in resisting digital control. Informed by Critical Theory, the conversation examines how algorithmic systems, rather than neutral tools, are caught in discursive formations that support dominant power relations, justifying surveillance, censorship, and political exclusion. At the same time, these very technologies create room for subversion, creativity, and collective resilience. The patterns it reveals illustrate the repressive logic of technological governance and the inventive tactics of civic actors working within and against digital authoritarianism.

Category A: Mechanisms of AI-Driven Control

One of the most salient and routine forms of repression comes from the penetration of AI into the lives of Iranians daily. Governments have repurposed hardware that was once designed for bureaucratic or municipal uses, such as clever traffic cameras, identification systems, or facial recognition on subways, to be used instead for controlling political opponents. “Surveillance and biometric technology is very much used to identify, track and terrify activists when it comes to protesting and when it comes to open challenge to what the state wants, be it hijab [compulsion],” activists and workers in different spheres said. As Critical Theory teaches us, this is a way in which technology can be used as a method of social control,



infusing surveillance into the texture of everyday life and creating a “chilling effect” that deters both political involvement and self-expression.

AI-based content moderation was another central theme, particularly for global-level platforms such as Instagram, Twitter, and Telegram. When I asked student protesters and activists, they pointed to how Persian-language political speech on the platform is frequently misunderstood by moderation algorithms, resulting in shadow bans, takedowns, and account suspensions. These machine-based systems, which have been trained on data primarily collected from the West, lack the cultural, social, or political understanding necessary to identify dissent from a non-Western country. It means that political speech is not just censored domestically, but it’s also censored transnationally. This is the algorithmic silencing of subaltern voices at work, entrenching the kind of epistemic injustice—one of the root insights that the Critical Theory tradition is predicated on—where the voices most integral to the resistance are structurally silenced from public discourse.

While AI may be the “way” the Chinese are producing state governance, AI technologies themselves are not inherently political inventions; they are neutral technologies that both democracies and autocracies have equal access to and can use to serve different ends. Interviewed experts stressed how systems like predictive policing and social graph analysis are turned into instruments of repression when they are employed to pre-emptively categorize critics as security threats. These tools and systems represent not only the motives of the people who use them but also the biases inherent in their training data and design decisions. In the Iranian case, describing political activism as extremism is supported by algorithmic design, legitimating state violence via a façade of ostensibly objective technology. For there, then, the criticism of instrumental reason by Critical Theory is crucial. The claim to neutrality wraps the reproduction of state ideology via code.

Even though Iran’s surveillance capabilities are advanced, the country has effectively no enforceable data protection or privacy rights. One very



concerning signpost on the road to this dystopia is the "User Protection and Core Online Services" (UPCOS) law, which, experts and civil society have argued, demonstrates the pattern of legal authoritarianism, where quasi-formal legal structures are put in place as a fig leaf for repression. This is touched upon, but the gap between universal ethical norms (such as OECD AI principles) and governance in authoritarian states is notable. As Critical Theory tells us, the law in such systems is often there not to contain power but to ritualize it - and to make the practice of power appear normal, just another day in the office, just a good day's work – as the most recent spectacle of legal principles trumping realpolitik returns to the inky darkness of impunity.

Category B: Resistance and Civic Counteractions

In the face of extensive surveillance, Iranian activists have adopted a variety of tactical practices to maintain anonymity and secure communication. These include turning off features like biometrics, the proliferation of digital literacy training, and using encrypted messaging apps (such as Signal and Briar) and burner phones. Activists emphasized the tools' practical utility, as well as the symbolic act of resistance they represent. However, these methods are limited by uneven access to resources, sanctions on vital technologies, and state control over the Internet. Such conditions highlight the precariousness of subversive activity within authoritarian contexts, where resistance is both an urgent modality and a creative response to constraint.

As the government increasingly deploys generative AI, biometric ecosystems, and smart city frameworks, activists and technologists are finding ways to hack back against these technologies. Catalyzed by the widespread adoption of tracking and meddling technologies, adversarial techniques for obfuscation and the open development of circumvention tools emerge as a grassroots civic tech movement. These projects aim to reclaim technology from authoritarians. Yet this resistance is frequently atomized and under-resourced, precarious to state violence as it is to international isolation. This tension is emblematic of the dialectic of innovation, in which



technologies of interest are deployed for both domination and resistance (Category B), and should be seen as an imperative for broader solidarities and institutional support for threatened civic tech practices.

7. Conclusion

7.1 Answers to research questions

Answers to research questions

Concerning the first research question, how does the authoritarian regime in Iran utilize artificial intelligence to repress civil society? I found AI to be a key element of the state's surveillance and control infrastructure. The regime employs a range of AI-based tools, including automatic license plate readers, predictive policing algorithms, facial and gait recognition systems, and natural language processing (NLP) tools, to track and suppress dissent. Though many of these systems were based on reverse-engineered Chinese and Russian models, Iran is increasingly producing them domestically, communicating not only a military-industrial strategy but also a long-term commitment to technological repression. Embedded in the smart city are AI-enabled tools used in security, urban management systems, telecommunications, and e-government services, which in turn enable surveillance to become routine. The process is also institutionalized through legislation, such as the User Protection and Core Online Services (UPCOS) bill and the National Information Network (NIN), which require platforms to comply with state surveillance and censorship demands. AI, therefore, doesn't function as a neutral technology but as a discrete, extended, and legally empowered tool of authoritarian political control.

For the second research question, the impact of AI on the behavior, communication, and security of civil activists in Iran, the findings reflect digital oppression in Iranian cyberspace. Meanwhile, in the streets, activists are closely monitored. During protests, demonstrators are tracked using artificial intelligence and facial recognition, as well as on-the-fly alert



mechanisms, to identify participants. These are the platforms that we fear are using AI moderation tools behind closed doors to delete activist content and silence dissent, namely to prevent it. Online platforms that use opaque AI moderation may frequently delete activism content and suspend accounts without explanation. Activists adapt by resorting to self-censorship, using VPNs and encrypted messaging applications, refraining from sensitive search queries, and shifting from online to offline mobilization. Others use coded language and throwaway “burner” phones to avoid detection. It goes deeper than tactical changes; this has a psychological hold. As one activist put it: “AI is not a tool for us; it is the closure of our act of seeing and our speaking. This quotation captures how AI not only configures the external terrain upon which activism plays out but is also profoundly implicated in activists’ sense of agency, modes of communication, and a sense of risk and insecurity when operating under autocratic regimes.

7.2 Achievements

This paper aims to explore the role of AI technologies in the context of autocratic regimes to monitor and suppress activist movements, particularly the Islamic Republic regime of Iran. From the critical theory perspective, coupled with the qualitative thematic methodology, the research demonstrates how AI-facilitated digital repression is constructed through intersections of technology, politics, and humans. Based on interviews with AI experts and activists, the findings emphasize that AI is not only a technological tool, but a profoundly political form of infrastructural control, bound up with ideology, governance, and resistance.

The themes at the core of this study from routine surveillance and algorithmic censorship, to resistance mechanisms and the ambiguities of AI as well as counter-technologies — reflect a complex and fast-changing landscape of digital authoritarianism. AI is deeply embedded in public life in Iran through facial recognition in traffic systems, biometric identity management and social media monitoring. These are instruments of



preemptive repression, especially against those who oppose gender norms, labor exploitation, ethnic discrimination, or environmental injustice.

The concentration of surveillance infrastructure, backed by legislation such as the User Protection and Core Online Services Bill centralizes state authority and closes down avenues for safe dissent. These infrastructural mechanisms reinforce how technological systems can codify political intent, especially when democratic safeguards are absent.

Importantly, this thesis problematizes the idea of AI as a neutral or general-purpose technology. In authoritarian systems like Iran, AI is not simply misapplied; it is intentionally weaponized for political repression. Current global AI ethics frameworks offer limited recourse, as they are predicated on democratic norms—transparency, contestation, and accountability—that do not exist in Iran. As a number of experts observed, ethical AI is impossible without enforceable rights and independent oversight.

This research also captures the creative resilience of activists subverting digital repression through encrypted communication, decentralized coordination, and grassroots digital literacy. However, such resistance is neither inevitable nor sustainable in the absence of support. It is fragile, under-resourced, and frequently criminalized. What is urgently required is international solidarity beyond rhetoric: Supporting local developers, defending digital rights, and holding tech companies and governments accountable for enabling repression.

This study contributes to the field of peace and development studies by bringing to light the political and repressive possibilities of AI in authoritarian conditions. It spans AI ethics, critical theory, and digital activism, offering both a warning and a call to arms. The next steps will involve applying this framework comparatively to other authoritarian contexts, exploring transnational AI surveillance dynamics—including corporate complicity—and investigating how alternative technological imaginaries can be rooted in grassroots contestations.



In sum, the fight over AI in Iran is more than code or algorithms—it is a fight over power, voice, and the right to resist. As AI becomes central not only to governance but also to the architecture of repression, the urgency of understanding and confronting its use in authoritarian contexts becomes all the more critical.



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Appendix

Interview guide

Questions from an AI expert
1. In your opinion, What are the core technical capabilities of AI that can assist the state in surveillance or control?
1.1: Can you give examples of when any of these capabilities have been used in authoritarian contexts?
1.2: In what ways are these different from the processes by which they are used in democratic governance?



1.3: What specific AI technologies or infrastructures do you believe the Iranian regime is using to monitor or suppress activists?

1.4: Are there any known implementations (e.g., facial recognition, online surveillance) within Iran that align with global trends in authoritarian AI use?

2 – To what extent do algorithmically-based decision-making (such as predictive policing, content moderation, etc.) forms of control facilitate the state's control over acts of activism?

2.1: Are these systems inherently political in their design, or do they become political in their deployment?

2.2: How might embedded biases in these systems influence civil liberties?

2.3: What risks do these systems pose to marginalized or politically active groups in Iran?

3- From a governance perspective, how well do current ethical frameworks (e.g., HCXAI) counter the instrumental uses of AI in authoritarian regimes?

3.1: Why do these frameworks often fall short in authoritarian contexts like Iran?

3.2: What deficiencies have you identified in existing policy or legal mechanisms to prevent abuse?

3.3: Will international norms or civic tech interventions be impervious to these authoritarian applications?



4. How do you foresee the evolution of AI regarding political control—particularly in contexts where transparency and accountability are limited?

4.1: Could any trends (e.g., smart cities, generative AI) exacerbate surveillance?

4.2: What resistance or counter-innovation is possible or is already emerging?

4.3: How might the state in Iran leverage technologies like smart city platforms or biometric databases?

questions from the activist



Have you personally seen or experienced the use of surveillance technologies (such as facial recognition, digital tracking, or social media monitoring) by Iranian authorities?

1.1: Do you think AI-based surveillance has increased in recent years? If so, how has this changed activist organizing?

1.2: Are there specific technologies or platforms that you or your peers intentionally avoid due to concerns about surveillance?

2. How has online content moderation — whether by Iranian authorities or platforms — affected your ability to express yourself freely or organize politically online?

2.1: Have you seen specific examples of content being removed or people's accounts blocked with little to no explanation?

2.2 Do you think AI is a factor in this kind of censorship? If so, how?

3. Do you know of any tools or strategies activists in Iran use to protect themselves from AI surveillance or digital repression?

3.1: In your experience, have any of these worked?

3.2 What could make such tools more accessible or effective for others?



4. As an expert, what do you think needs to happen—technically, politically, or internationally—to mitigate the risks posed by AI to civil liberties in Iran?

4.1: Do you see any legal or policy frameworks, local/global, that may afford greater protection?

4.2: What role can international tech companies, human rights organizations, or governments play?

Consent Form for Participation in Research Interview

Title of Study: Smart Control: The Role of AI in Authoritarian Governance

Researcher: Mahdieh Golroo, Linnæus University

Purpose of the Study:

This study examines how artificial intelligence (AI) enables authoritarian regime(s) to use surveillance to suppress civic activism, with a special focus on the Islamic Republic of Iran. Centering on interviews with AI experts and activists, the research examines the technologies and tactics deployed and considers the pathways available for resisting.

Participation and Time at Will:

Your participation in this interview is strictly voluntary. The interview will be approximately 45–60 min and will be semi-structured. You can decline to answer any question or end the interview anytime, as in a face-to-face conversation, with no explanation. Your name and any identifying



information will remain confidential. All data will be made anonymous and kept confidential. Audio recordings (if you consent) and transcripts will only be used for academic purposes, and none of them will be shared with a third party.

Potential Risks and Benefits:

As its somewhat touchy subjects so far make possible, participants may feel uncomfortable, but in few circumstances should someone feel vulnerable discussing a specific episode of surveillance or repression. Protect your privacy. Your trust and safety are very important to us! Participation in the study does not involve any direct financial reward, but your perspective will help to inform important academic research on human rights and AI governance.

Your Rights:

You can leave anytime if you don't want to continue participating in the study.

You have the right to refuse to answer any question.

If something is unclear to you, or if you have concerns, you may ask.

You can ask for portions of your interview not to be quoted directly.

Use of Data:

The material will be applied for a master's thesis submitted to Linnaeus University and could be used in forthcoming academic papers or conference presentations. You will not be identified in any reporting of the study.

1 Consent:



Please check the boxes below:

- ☐ I have read and understood the purpose and nature of the interview.
- ☐ I understand that participation is voluntary and that I can withdraw anytime.
- ☐ I consent to participate in the interview.
- ☐ I consent to the audio recording of the interview (optional).
- ☐ I do not consent to audio recording but agree to note-taking.

Participant's Signature: _____

Date: _____

Researcher's Signature: _____

Date: _____